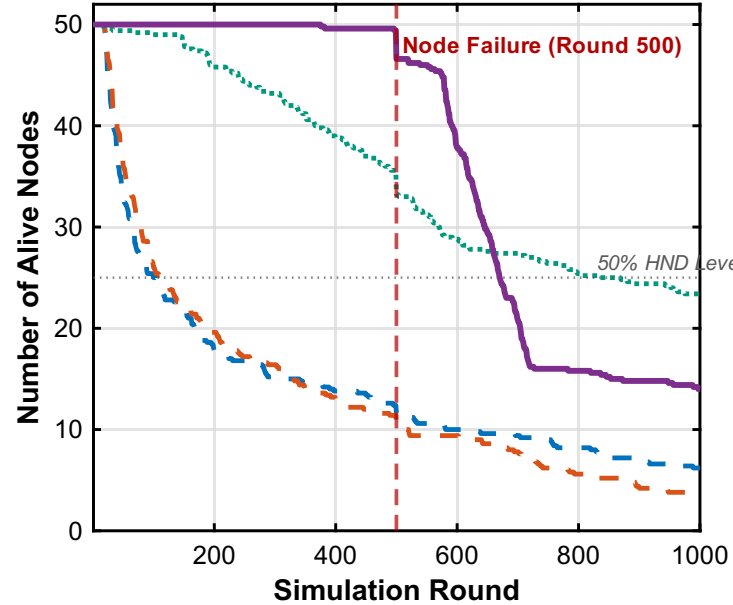


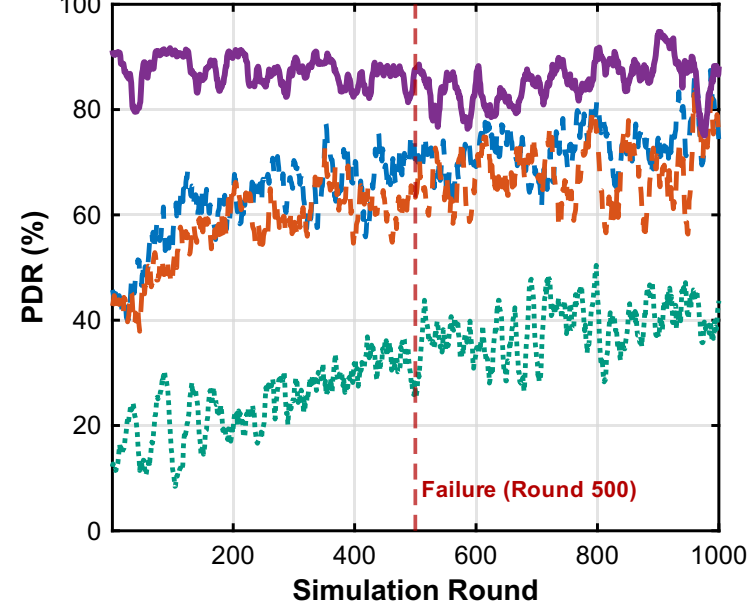
RL-Hybrid vs Baseline Protocols (LEACH, DEEC, PEGASIS)

Validated 5-Seed Trajectory Dashboard (Seeds 42, 43, 44, 45, 46 | 1000 Rounds)

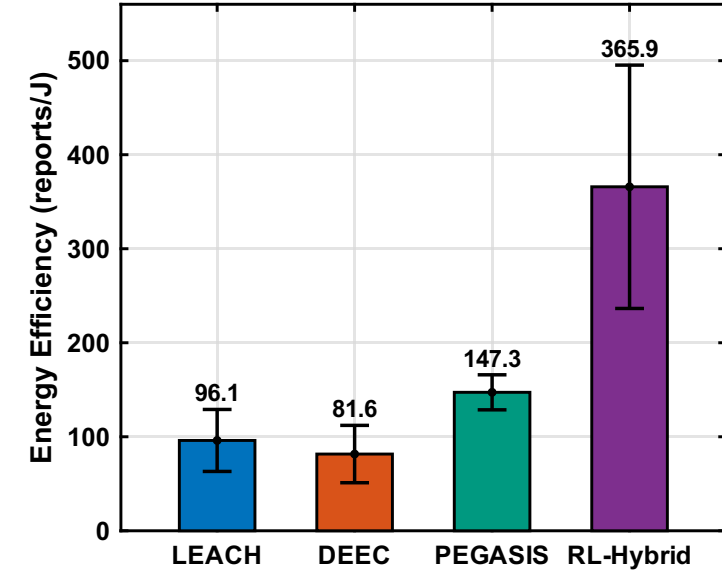
Network Lifetime — Alive Nodes vs Simulation Round



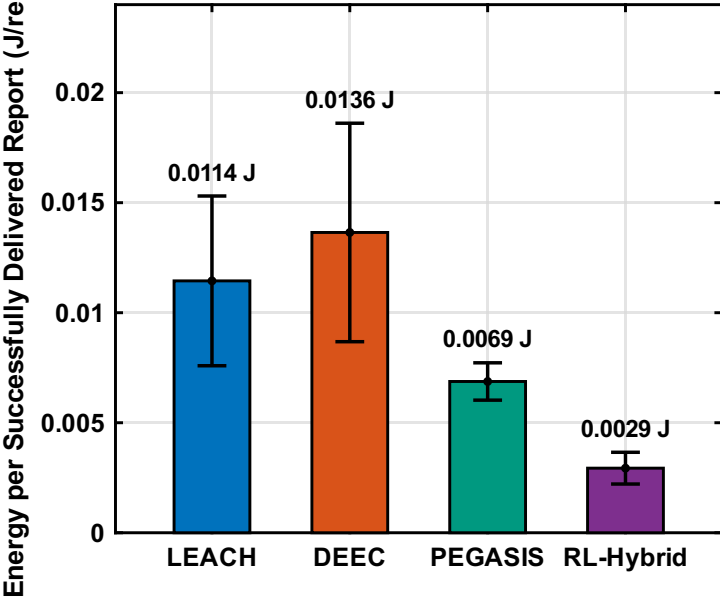
Packet Delivery Ratio



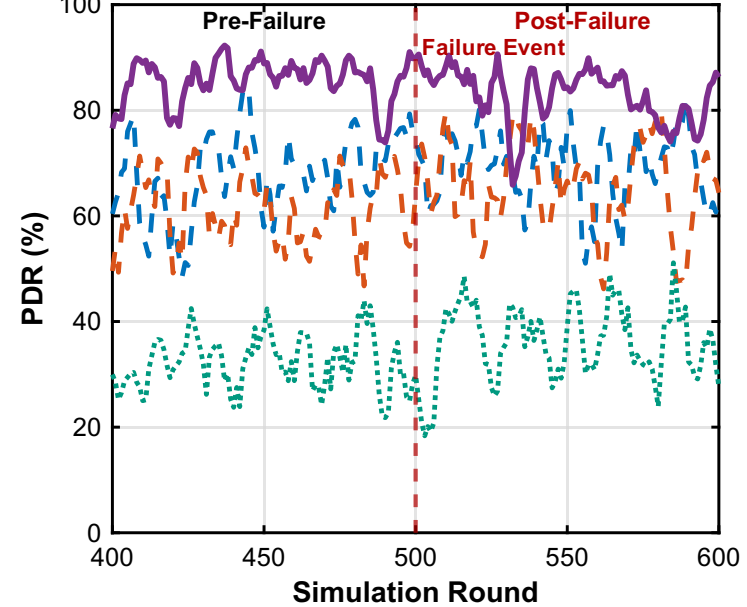
Final Energy Efficiency



Energy Cost per Successfully Delivered Report



Failure Recovery and Route Adaptation



PROTOCOL LEGEND

- LEACH
- DEEC
- PEGASIS
- RL-Hybrid (Proposed)
- Failure Event (Round 500)